

PortInput: Enabling Always-Available Micro-Gesture Input With Pressure Array Sensor

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Abstract—Micro-gestures provide a natural, efficient, and privacy-preserving input modality; however, existing techniques often depend on environmental conditions, limiting their robustness and applicability in real-world settings. In this work, we present three portable prototypes—finger-cot, finger-worn, and surface-based—that integrate compact pressure array sensors to support environment-independent micro-gesture interaction. We further propose a deep learning-based recognition model that accurately classifies 14 micro-gestures by analyzing temporal pressure patterns. Building upon these components, we introduce *PortInput*, a real-time interactive system that enables robust micro-gesture tracking and detection. We conducted two user studies with augmented reality (AR) head-mounted displays (HMDs). The first study evaluates input performance under both *sitting* and *walking* conditions, while the second compares *PortInput* with a commercial pressure-based ring device. The results show that *PortInput* improves usability and user experience, achieves comparable accuracy, and enables faster input with lower perceived workload. Overall, *PortInput* have potential to offer efficient, robust, and comfortable input across diverse application scenarios—ranging from AR/Virtual Reality (VR) headsets to smart homes and in-car systems—even in noisy or cluttered environments. This work provides a foundation for integrating pressure array sensors into ring-based or other portable devices, advancing always-available micro-gesture interaction for ubiquitous computing environments.

Index Terms—Wearable computing, ubiquitous and mobile devices, human-centered computing, gestural input, interaction techniques, human-computer interaction, head-mounted displays.

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I. INTRODUCTION

THIS Micro-gesture interaction [1] enables natural and intuitive input through subtle finger movements such as tapping, sliding, and circling. It requires minimal physical effort, supports prolonged use, and can be performed discreetly, improving privacy and social acceptability. As micro-gestures align well with everyday motor behaviors, they reduce cognitive and learning burdens compared to more complex input techniques [2]. Consequently, micro-gesture interaction is well suited for a wide range of mobile and wearable platforms, including smartphones, smartwatches, smart glasses, and AR/VR HMDs.

Existing studies have investigated micro-gesture detection using a wide range of sensors, including cameras [2], [3], acoustic sensors [4], [5], [6], [7], [8], infrared and magnetic sensors [9], [10], [11], electric field sensors [12], and inertial measurement units (IMUs) [13], [14]. While effective, many of these approaches are susceptible to environmental interference, which limits their robustness and scalability in applications. This limitation becomes more pronounced as AR/VR HMDs are increasingly used across diverse and dynamic environments. In contrast, pressure sensors are inherently robust because they respond directly to user-applied forces and are largely unaffected by external environmental conditions [15], [16]. Prior work has demonstrated the feasibility of using pressure sensing for recognizing micro-gestures such as clicks and swipes [16], [17], [18], and recent commercial ring devices have begun adopting pressure sensors primarily for click-based input [19], [20].

Therefore, we integrate a pressure array sensor into a compact ring-shaped and miniature portable form factor. Compared to ring- or surface-based systems relying on other sensing modalities, such as IMUs [21] or acoustic sensors [5], our approach provides more stable and reliable interaction across diverse environments. Moreover, the high spatial resolution of the pressure array enables users to perform rich micro-gestures using subtle thumb or forefinger movements on a small interactive surface. Compared to existing pressure-based devices [16], [17], our design supports a broader set of micro-gestures without requiring large or conspicuous motions, thereby improving social acceptability in public settings. Furthermore, by leveraging users' proprioception, these devices support eye-free micro-gesture input, which enhances usability and ensures user safety across various environments. Users can perform natural and robust interactions in both stationary [6], [22] and mobile scenarios [23], [24].



Fig. 1. A real-time gesture detection system, *PortInput*, that supports robust micro-gesture interaction.

Inspired by existing gesture-enabled devices, we have designed three pressure-sensor-based prototypes positioned at common interaction locations: *finger-cot*, *finger-worn*, and *surface-based* prototype. These prototypes enable micro-gesture input at the fingertip [25], [26], along the side of the index finger [23], [27], [28], and on stationary surfaces such as desks or armrests [6], [22]. We then review existing micro-gesture studies and summarize a set of 14 representative micro-gestures, for which we develop a deep neural network-based recognition model capable of classifying them using temporal sequences of pressure data. The model is evaluated comparatively with existing neural network algorithms and evaluated in both within-user and across-user settings. Experimental results demonstrate high within-user classification accuracy across all locations (Chair: 98.35%, Table: 98.89%, Fingertip: 95.5%, Finger: 98.26%) and robust generalization in cross-user scenarios (Chair: 94.14%, Table: 96.79%, Fingertip: 91.84%, Finger: 97.76%).

Building upon our prototypes and recognition model, we implement *PortInput*, a real-time gesture recognition system that supports robust micro-gesture interaction (Fig. 1). We evaluate its effectiveness through two user studies on an AR HMD. The first study includes an icon selection task, comparing *PortInput* with two widely adopted mid-air interaction techniques, *Pointing-Pinch* and *Air-Tap*, under both *sitting* and *walking* conditions. The results show that *PortInput* achieves good system usability, reduced physical demand, and improved user experience, particularly in mobile scenarios. To further assess its practical applicability, a second study compares *PortInput* with a commercial pressure-based ring device in flat and hierarchical menu selection tasks. The results indicate comparable accuracy but significantly faster task completion, lower perceived workload, and higher subjective usability and user experience for *PortInput*. Overall, these studies demonstrate the feasibility of using compact pressure sensors for reliable, eyes-free micro-gesture input and highlight their potential for integration into ring-based or other portable devices to support always-available, environment-independent interaction. There are three-fold primary contributions:

- 1) We propose *PortInput*, a portable input prototype equipped with a pressure sensor array that can be either placed on

arbitrary surfaces or worn on the user's index finger. By continuously capturing the temporal pressure patterns in real time, *PortInput* has the potential to support eye-free and ubiquitous micro-gesture interaction.

- 2) We develop a deep neural network-based gesture recognition algorithm capable of identifying 14 distinct micro-gestures performed at various interaction positions.
- 3) We conduct two user studies to evaluate *PortInput*. The first study examines its input performance under both *sitting* and *walking* conditions in comparison with commonly used mid-air interaction techniques in AR HMDs, while the second study compares *PortInput* with a commercial pressure-based ring device in flat and hierarchical AR menu selection tasks, demonstrating its efficiency, usability, and practical applicability.

II. RELATED WORK

Our goal is to develop a portable system for micro-gesture input, leveraging a pressure array sensor to support interaction in both stationary and mobile scenarios, particularly in mobile HMDs. This work is inspired by prior research on touch-based gesture interaction AR/VR, micro-gesture interaction enabled by wearable devices, and pressure-based input techniques.

A. Touch Gesture Interaction for Mobile HMDs

Utilizing the human body, hands, and other surrounding surfaces as an input medium enables seamless, convenient, and ubiquitous input. Previous studies have explored gesture input through touching various body parts [29], [30], [31], [32], [33] and thumb-to-hand contact [16], [25], [26], [34], as well as on-surface touch [5], [21], [35], [36] for input in Mobile HMDs. For example, Wang et al. [29] developed a wrist-mounted prototype to detect multi-stroke drawing-based gestures on palms by using an IR camera and laser-line projector, where the user could sketch on the left palm by right index finger. Meier et al. [35] introduced TapID, a wrist-based inertial sensing system that integrates a pair of inertial sensors into a flexible strap, with one sensor positioned on each side of the wrist. This system could detect surface tapping made by different fingers to trigger

inputs in environments by complementing headset-tracked hand poses. Lissermann et al. [30] introduced EarPut, a prototype worn behind the ear, to detect the touch behaviour in different areas in the ear for input, including swipe, single and multi-touch gestures from fingers. Liu et al. [5] introduced AudioGest, a portable small device that could detect the audio signal generated by finger tapping and sliding on different surfaces through a set of microphone devices without extensive calibration, enabling finger-driven tap, double tap, triple tap, slide up/right/left/down gesture recognition for HMD. Xu et al. introduced a device with a capacitive sensor to detect the one-hand and both-hand thumb-to-fingertip touch input for text entry [25], [26]. Chen et al. [7] introduced GestOnHMD, which could detect the tapping and the scratching gestures on the front, the left, and the right surfaces on a mobile HMD.

Our system extends existing input methods by enabling users to perform gesture input anytime and anywhere, even while walking. This expands the scope of interaction beyond the special part of the body or specialized surfaces, enhancing flexibility and enabling more versatile use in mobile conditions.

B. Micro-Gesture Interaction With Portable Devices

One of our goals is to enable users to perform precise micro-gesture input across diverse environments. Portable and wearable devices play a crucial role in supporting this goal, and they have been studied for micro-gesture interaction.

These devices include wrist-worn devices [14], [37], finger-worn rings [1], [8], [21], [23], [38], [39], [40], combinations of wristbands and rings [41], finger-cot devices [25], [26], nail-mounted systems [10], [11], [13], [42], and ear-worn devices [4], [30]. Additionally, some systems are designed to be placed directly on environmental surfaces to support surface-based interaction [3], [5], [6]. These devices leverage different sensors to detect micro-gestures, including cameras [2], [3], acoustic sensors [4], [5], [6], [7], [8], infrared proximity sensors [9], photo-reflective sensors [42], [43], magnetic sensors [10], [11], electric field sensors [12], bio-acoustic sensors [44], radio-frequency sensors [37], electromagnetic coupling sensors [41], contact conductivity sensors [34], and inertial measurement units (IMUs) [13], [14]. For example, Boldu et al. [23] developed a ring device with an integrated touchpad PCB capable of recognizing four swiping gestures and tapping, and evaluated its performance under standing, walking, and running conditions. Bardot [45] introduced a capacitive ring-based prototype supporting on-ring, on-object, and in-air interaction for tapping, flicking, and gesture input. Gong et al. [39] proposed Pyro, a thumb-tip gesture recognition device using pyroelectric infrared sensing to detect shapes such as triangles, checkmarks, rectangles, circles, and finger rubs. DualRing [21] captured hand and finger states to support gestures including pinching, swiping, and tapping. Hanayama et al. [43] developed SkinRing, a ring-shaped device with eight photo-reflective sensors enabling side-finger gesture input, including directional swipes, taps, double taps, check marks, and clockwise/counterclockwise circles.

Compared to other sensors, pressure sensors are less affected by environmental factors such as lighting conditions [3],

magnetic interference [10], surface material [5], and hand orientation [46]. They are also capable of detecting fine-grained micro-gestures with higher accuracy. Our system leverages the pressure array sensor, that can be deployed both on the user's finger or on surrounding surfaces to support robust and portable micro-gesture interaction.

C. Pressure Sensor-Based Input

Pressure sensors offer both high portability and sensitivity, making them well-suited for hand gesture [49], [50], [51], [52] and micro-gesture recognition tasks [16], [17], [18], [48], [49], demonstrating the feasibility of pressure-based gesture input.

Commercial products have already integrated pressure sensors into ring-based interactive devices, such as the FinchRing [19], NOLO RING [20] and Genius Ring [47], which support basic touch-based interactions on the ring surface. Previous studies have also used pressure sensors for interactions such as cursor control and micro-gestures, as shown in Table I. For example, Kubo et al. [48] introduced a ring-type pointing device utilizing a three-axis pressure sensor to control a 2D cursor on a large display. Jiang et al. [16] developed HiFinger, a glove-based system with six pressure sensors positioned on the inner side of the index and middle fingers, allowing users to input text through thumb tapping. Lim et al. [17] presented OctaRing, an octagon-shaped ring with multiple pressure sensors capable of recognizing multitouch gestures involving one to four fingers. Wang et al. [18] employed a pressure-sensitive ring to detect sliding gestures based on pressure variations across different regions, aimed at enhancing gaze-based input. However, their study did not include a quantitative evaluation of gesture recognition accuracy, limiting the assessment of its performance.

In our work, we propose the use of a compact pressure sensor array to support the recognition of fine-grained micro-gestures such as tapping, sliding, and circling performed by the thumb and index finger, expanding the number of micro-gestures supported by existing pressure sensor-based recognition methods. Our system would also support being worn on the index finger or placed directly in the environment, facilitating use in a wide range of scenarios.

III. PROTOTYPE DESIGN

A. Design Consideration

Comfortable: In our work, we explore three distinct micro-gesture input locations tailored to different application scenarios and user preferences based on previous work [13], [25], [27], [45]: (1) the index fingertip, (2) a ring worn on the index finger, and (3) surrounding surfaces in the environment. Thumb-to-index fingertip gestures, such as taps and strokes, have been demonstrated to be more comfortable than gestures involving other finger combinations [11]. Furthermore, wearing a ring on the proximal phalanx of the index finger has been identified as a common and comfortable configuration [27]. Based on these findings, we propose a finger cot-based design that supports micro-gesture input from the thumb to the index fingertip. The cot integrates additional components worn in the form of a ring

TABLE I

SUMMARY OF REPRESENTATIVE PRESSURE-BASED INTERACTION. THE SENSOR LATENCY IN ALL SYSTEMS IS IN THE MILLISECOND RANGE AND HAS NO IMPACT ON THE INTERACTION.

System	Form	Interaction	Accuracy (Range)	Touch specs
FinchRing [19]	Ring	-	-	-
Genius Ring [47]	Ring	Cursor control	-	-
NOLO Ring [20]	Ring	Cursor control	-	12mm*13mm
Kubo et al. [48]	Ring	Cursor control	-	-
HiFinger [16]	Finger-cot	Click	0.01N (0.2-10N)	10mm circle
OctaRing [17]	Ring	Click	- (0.2-20N)	5mm circle
Wang et al. [18]	Ring	Micro-gestures	-	14mm*14mm
Our	Ring; finger-cot; surface-based	Micro-gestures	0.01N (0-30N)	14mm*14mm

to enhance functionality while maintaining comfort. Previous studies also suggest that thumb-to-ring surface interactions offer a natural and ergonomic input method [45]. Consequently, we incorporate this interaction method into our design by enabling micro-gesture input through thumb contact with the ring surface. Moreover, when users are in a fixed position with access to a stable support surface, resting the hand and performing gestures with the index fingertip on that surface has been shown to offer higher comfort compared to mid-air gestures [13]. To accommodate such use cases, we extend our input design to support micro-gesture interaction on fixed surfaces such as tables or chair armrests.

Eyes free: We aim for our system to be usable in diverse environments, including while walking. In many scenarios, particularly for safety and convenience, users must input commands without relying on visual attention. To support such eyes-free interaction, we design micro-gestures such as tapping and sliding. These gestures can be performed using proprioception, either by tapping the fingertip or ring with the thumb, or by clicking the device surface with the index finger, without the need for visual guidance on the touch surface. Pressure sensors enable robust and accurate detection of these micro-gestures. Furthermore, compared with mid-air gestures, pressure-based input provides tactile feedback through physical contact, helping users perceive gesture completion via touch and enhancing the reliability of eyes-free interaction.

Social acceptability: Compared to other hand-worn input devices, ring-like form devices are generally more socially acceptable [53]. Such devices are compact, easily removable, and well-suited for thumb and fingertip-based interaction, allowing users to maintain freedom of hand movement. We adopt a ring-like form that enables micro-gesture input on a compact surface—either directly on the ring or on the fingertip via thumb interaction. Additionally, in stationary scenarios, the device can be placed on a fixed surface, enabling gesture input using the index finger. Both approaches minimize conspicuous movements by leveraging subtle fingertip gestures, thereby reducing social attention and increasing user comfort in public.

B. Interaction Scenario

With the advancement of mobile HMDs, users increasingly engage with these devices across diverse environments—not only while seated but also while walking [22], [23], [24]. To support such ubiquitous use, our system leverages pressure

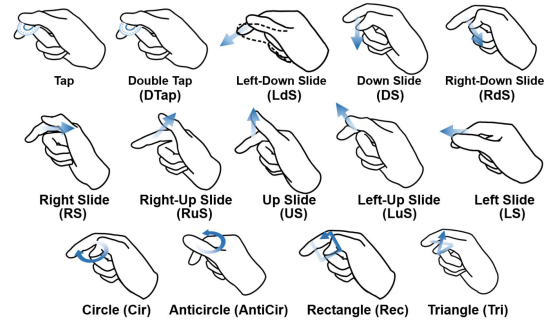


Fig. 2. 14 micro-gestures. For directional slide gestures, sliding outward along the index finger's middle phalanx corresponds to the Right Slide, while sliding perpendicular toward the palm corresponds to the Down Slide.

sensors to mitigate environmental interference and enable robust gesture recognition across varying contexts, including both mobile and stationary scenarios. In seated settings, surfaces such as desks and chair armrests are commonly utilized for interaction [5], [22]. Therefore, we have considered the user's interaction with the surrounding surfaces, such as the table and chair armrest.

C. Gesture Mode

Previous research has proposed various micro-gestures for interaction [1], [2], [5], [23], [39], [42], [54]. Based on these studies, we compile an initial set of candidate gestures and ultimately select a gesture set consisting of 14 micro-gestures, categorized as follows: **Tap Modes:** We select *Tap* and *Double Tap* gestures, which are commonly used in prior work [1], [2], [5], [23], [42]. **Directional Slides:** Previous studies have typically employed four-directional sliding gestures (i.e., up, down, left, and right) [1], [2], [5], [21], [23], [42]. Leveraging the high spatial resolution of our pressure sensor array, we extend this to eight directions: up (US), down (DS), left (LS), right (RS), left-up (LuS), left-down (LdS), right-down (RdS), and right-up (RuS). **Continuous Gestures:** We also include more complex, continuous gesture patterns, such as *Triangle*, *Rectangle*, *Circle Clockwise* (Cir), and *Circle Counterclockwise* (AntiCir) [1], [39], [42]. In summary, the final gesture set includes: Tap, Double Tap (DTap), eight directional slides (US, DS, LS, RS, LuS, LdS, RdS, RuS), and four continuous gestures (Cir, AntiCir, Rec, Tri), as illustrated in Fig. 2.

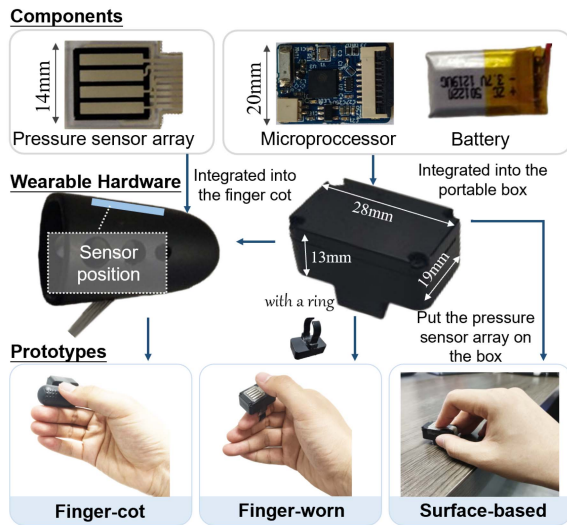


Fig. 3. Three different prototypes.

D. Prototype

1) *Prototypes for Different Situations:* In this work, we have developed three types of prototypes tailored for different interaction contexts: a finger-cot prototype, a finger-worn prototype, and a surface-based prototype, as shown in Fig. 3.

Finger-cot Prototype: Since humans can perform fine-grained interactions between the thumb and other fingertips using proprioception, many previous studies have explored micro-gesture input on the index finger [25], [26], [39], [42]. Therefore, we have implemented a finger-cot-based prototype, using a silicone shell with 40HA hardness as shown in Fig. 3. The pressure array sensor can be embedded in finger-cot and fixed to the index fingertip of the user, thereby supporting micro-gestures from the user's thumb to the index fingertip.

Finger-worn prototype: Many studies have also focused on micro-gestures from the thumb to the side of the index finger, which are divided into direct thumb-to-finger and thumb-to-ring gestures. Thumb-to-finger [21], [22], [43], [54] usually uses sensors for micro-gesture detection, such as cameras, photo-reflective sensors, acoustic sensors, and IMUs, which are not easily affected by environmental factors such as light, noise, and electromagnetic fields. For the thumb-to-ring gesture, the user directly interacts with sensors [23], [24], [45] such as capacitive sensors on the ring worn on the index finger, which is not easily affected by environmental factors. Therefore, we implement a Finger-worn prototype and thumb-to-ring gesture based on a pressure sensor, which supports users to achieve robust micro-gestures in different environments.

Surface-based prototype: Previous studies have implemented micro-gestures on different interfaces such as desktops and walls with sensors such as sound, cameras, and IMUs, expanding the scope of interactive applications [5], [13], [55]. However, the above devices are easily affected by environmental factors such as noise and light or need to be worn all the time. Therefore, we extend the above finger-worn prototype and propose a surface-based prototype, which allows users to place the device on any

surface in the environment and implement micro-gesture interactions based on pressure sensors based on human orientation perception.

2) *Hardware:* All prototypes share a common hardware configuration, consisting of a pressure array sensor, a microprocessor, a battery, and a Bluetooth communication module. The pressure array sensor (M0404S piezoresistive flexible pressure sensor) measures $14\text{ mm} \times 14\text{ mm}$ and supports a sampling rate of 60 Hz. It comprises 16 resistive pressure sensors arranged in a 4×4 grid, with an inter-sensor spacing of 0.8 mm. Each sensor unit can detect a pressure range from 0 to 30 N (accuracy 0.01N) with digital output from 0 to 3000 (accuracy 1) and response time $< 20\text{ ms}$. The microprocessor, based on the ESP32 chip, receives analog signals from the sensor array, computes the pressure amplitude of each sensor unit, converts the pressure values into digital signals, and transmits them to a host computer via Bluetooth. The microprocessor, battery, and Bluetooth module are all integrated into a compact housing measuring $28\text{ mm} \times 19\text{ mm} \times 13\text{ mm}$, which is lightweight and convenient for wearable use on the finger. Specifically, in the **Finger-cot prototype**, we fabricate a finger sleeve using silicone material and embed the pressure sensor array at the fingertip area to support thumb-to-finger interactions (Fig. 3). In the **Finger-worn prototype**, the sensor array is mounted on a ring structure, allowing users to wear it on the side of the index finger. In both the **Finger-worn** and **Surface-based prototypes**, the pressure sensor is affixed to the top surface of the device housing box, enabling users to perform micro-gesture interactions directly on the device surface (Fig. 3).

IV. DEEP LEARNING-BASED GESTURE CLASSIFICATION

A. Data Collection

1) *Participants and Apparatus:* We recruited 20 participants aged 21-28 years (12 males, 8 females, $M = 25.37$, $SD = 4.28$) for data collection. All participants are right-handed and performed all tasks using their right hand. The experiment has obtained the informed consent of the participants. The experiment is conducted on a desktop computer running Windows 10, equipped with an Intel Core i7, 16 GB of RAM, and an NVIDIA RTX 3070Ti. The software is developed using C# in Unity 2022.3.45f1c1. A Dell SP2318H monitor (23.6-inch, resolution of 1920×1080 pixels) is used to present the visual interface during the experiment.

2) *Task and Procedure:* Participants perform micro-gestures at four **positions** (**Fingertip**, **Finger**, **Table**, **Chair** armrest; Fig. 4) using the corresponding prototypes (four tasks). All tasks are identical except for the location and device, and task order was counterbalanced using a Latin square. Participants are given a short training session to familiarize themselves with the gestures before formal data collection begins. Each task lasts 15 minutes with 5-minute breaks, and the entire experiment takes 80 minutes.

In each task, participants perform micro-gestures on the prototype in response to a randomized micro-gesture displayed on a display as shown in Fig. 5. A micro-gesture is shown on the display, and participants are instructed to perform the displayed

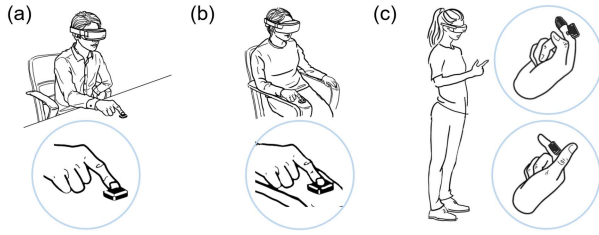


Fig. 4. Four different positions where the user performs the micro-gestures. (a) Table; (b) Chair armrest; (c) Fingertip (top) and Finger (bottom).

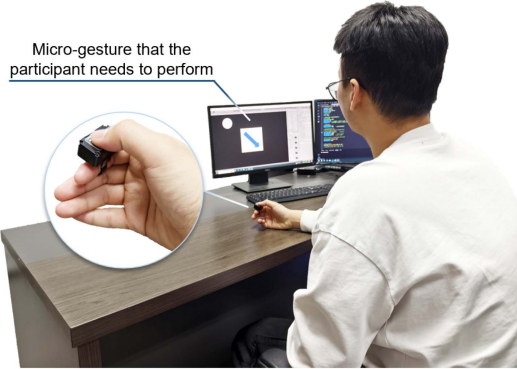


Fig. 5. Illustration of the experimental setup for data collection. The figure shows the collection of the Finger group, and the same settings are used in the other groups. The participant performs corresponding actions according to the micro-gestures shown in the display.

gesture, during which interaction data from the pressure sensor array are recorded in real time. This sensor data is labeled according to the corresponding gesture and later used for model training. Each micro-gesture is shown on the display and remains visible until the participant completes the gesture and presses the space bar to proceed to the next micro-gesture. In addition, if a user performs a clearly incorrect gesture (e.g., the gesture type or direction is incorrect), the user presses the "delete" key, and that segment of data is marked and deleted during data preprocessing. Therefore, the final dataset only contains correctly performed gesture samples.

In total, we collect data of 14 (micro-gestures) $\times$ 20 (participants) $\times$ 20 (repeated) $\times$ 4 (positions) = 22,400 trials.

B. Data Processing and Representation

In micro-gesture data processing, data segmentation is initially performed based on the operational states of the pressure array sensor. During the active state (when users execute micro-gestures), more than 5 out of the 16 sensor channels consistently detect valid pressure signals, while in the idle State (non-execution periods), mechanical deformation introduces minor noise signals (with ≤ 3 active channels). Valid micro-gesture segments are extracted by defining initiation and termination markers as intervals in which ≥ 4 channels exceed a pressure threshold of 0.2 N for at least 3 consecutive frames. To preserve temporal integrity in double-click signals, all transitional data

between consecutive triggers is retained. We observed that the interval between users' double-clicks was less than 0.3 seconds, which aligns with the typical double-click interval (0.2–0.5 seconds) found in mobile phone touchscreens, computer touchpads, and similar devices. For all micro-gestures, natural inter-user variations in execution speed, amplitude, and trajectory were observed. Therefore, we applied a time-aligned representation to normalize gesture sequences of different durations to a uniform length while preserving their temporal characteristics, and combined it with data augmentation to enhance robustness to user-dependent variations. Sequences shorter than N frames are extended using cubic spline interpolation, while longer sequences are downsampled to N frames via average pooling with a stride of 2. Maximum-minimum normalization is then applied to map channel data into the interval $[0, 1]$. Finally, to enhance model robustness against noise interference, we apply parameterized Gaussian noise injection defined as: $\sigma = 0.02 \times (x_{\max} - x_{\min})$ where $x_{\max}$ and $x_{\min}$ denote the maximum and minimum pressure values detected within the respective micro-gesture. This process, scaled by the dynamic signal range, is used to augment the dataset by a factor of three while maintaining signal fidelity.

We represent each frame of data collected by the 16 pressure sensor channels at time t as: $x_t = [p_t^1, p_t^2, \dots, p_t^{15}, p_t^{16}]^T$ where p_t^n represents the measured value from the n -th channel at frame t . A complete micro-gesture data is denoted as: $X = \{x_t\}_{t=0}^N \in \mathbb{R}^{16 \times N}$. In our experiments, we set $N = 64$ for time alignment of all gesture sequences. Let g denote the identified micro-gesture type, and f_θ the prediction model, where θ represents the parameters of the deep learning network. Then the performed micro-gesture can be inferred as:

$$g = f_\theta(X)$$

C. Recognition Model

We propose a deep learning-based micro-gesture recognition model that predicts the category of micro-gestures performed by users based on pre-processed time-series data collected from the pressure array sensor. During real-time detection, once gesture execution data is captured, the model is able to classify the micro-gesture performed by the user in real time.

Model: Our model is developed based on the VGG backbone and comprises multiple convolutional blocks designed to progressively extract hierarchical features while increasing channel depth, as illustrated in Fig. 6. At the input, the model changed from accepting the original 2D image input to receiving multi-channel time-series data of size $B \times 16 \times 64$, where 16 represents the number of pressure sensor channels, and 64 is the time-normalized frame length. Accordingly, the 2D convolution and pooling operations in the network were replaced with 1D convolution and pooling along the time dimension to model pressure changes during micro-gesture execution, treating sensor channels as feature channels. At the output, the original high-dimensional fully connected classification head of VGG for ImageNet is replaced with a lightweight fully connected layer, which outputs a one-hot probability vector for the micro-gesture category. Each convolutional block includes two convolutional

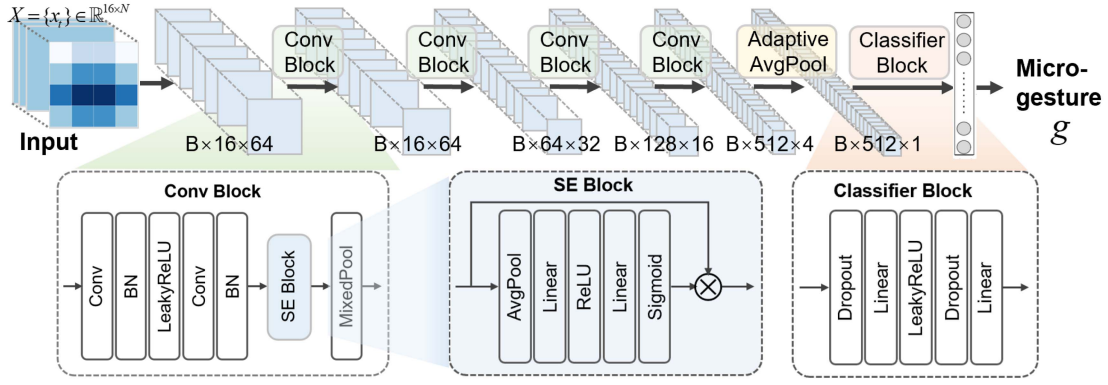


Fig. 6. Micro-gesture recognition network structure.

layers, followed by batch normalization and a LeakyReLU activation function to prevent neuron inactivation. This is adapted to process multi-channel pressure time-series data. To enhance feature extraction, we incorporate a channel attention mechanism using the Squeeze-and-Excitation (SE) block. Furthermore, a hybrid pooling strategy is applied after each convolutional block, combining the complementary strengths of average pooling (AvgPool) and max pooling (MaxPool) to retain richer feature information. An adaptive average pooling layer is then employed to compress the temporal dimension globally. The resulting global feature vector is passed through a fully connected classifier with dropout regularization for final micro-gesture classification.

Training: All models, including compared models, are trained on a laptop equipped with an NVIDIA RTX 4060 GPU, 32 GB of RAM, and an Intel i7-14650HX CPU. We adopt the Adam optimizer with an initial learning rate of 2×10^{-4} . The learning rate is decayed by a factor of 0.5 every 5 epochs to facilitate convergence and improve training stability. Each model is trained for 24 epochs with a batch size of 32. The entire implementation is developed using Python 3.9.20 and PyTorch 2.6.0+cu118.

D. Offline Performance

1) Data Usage: To evaluate the generalizability of micro-gesture recognition models, we implemented two data splitting strategies. 1) For within-user evaluation, data from 16 participants were used, with a 7:3 split for training and testing, respectively. 2) For cross-user evaluation, the models were trained on the data from the aforementioned 16 participants and tested on data from four leave-out participants.

2) Compared Models: Our dataset can be processed and classified using a range of well-established deep learning models. Therefore, we compared our proposed model with several representative architectures, including MLP-based, ResNet-based [56], DenseNet-based [57], VGGNet-based [58], Transformer-based [59], and CNN-LSTM-based models [60]. In addition, we added pressure variation features (dF/dt) to verify whether it could contribute to classification accuracy. Notably, the VGGNet-based model is implemented using the same convolutional backbone as our proposed architecture but without the

SE blocks, serving as an ablation baseline to explicitly evaluate the contribution of channel attention. We trained these models using the same training method as our model to ensure a fair comparison.

3) Results: Compare with other models: Table II presents the testing accuracy of each model under within-user and cross-user evaluations across four positions. Our proposed model consistently outperforms all baselines, achieving an average accuracy above 91.84% across all positions. Generally, within-user accuracy is higher than cross-user accuracy for all models. However, the cross-user accuracy drop in the Table and Finger groups is less significant compared to the Chair and Fingertip groups, likely due to lower noise levels in the Table and Finger datasets. These findings substantiate the effectiveness and robustness of our model in micro-gesture recognition. In addition, although additional features such as dF/dt were explored, they did not lead to consistent performance improvements. This is likely because temporal pressure variations are already captured by the convolutional filters operating across the time dimension. Explicit derivative features may introduce redundancy when the network has sufficient capacity to learn temporal dynamics directly from raw pressure sequences.

We further analyze the performance of our prediction model under both within-user and cross-user settings. **Within-user performance:** Fig. 7 presents the average recognition accuracy for each micro-gesture in the within-user test. All 14 micro-gesture classes achieve high precision (all above 95%), with per-gesture accuracies exceeding 91% across all interaction groups. Several micro-gestures reach perfect recognition in specific groups, particularly in the Table and Chair conditions. Despite the overall high accuracy, consistent confusion patterns are observed. The Rec gesture is frequently misclassified as Cir, while Cir is also prone to confusion with other gestures. Slide gestures in adjacent directions are often confused (e.g., LS vs. LuS), and DTap is commonly misclassified as Tap, especially in the Finger and Fingertip groups. **Cross-user performance.** Fig. 8 shows the corresponding results for the cross-user test. All micro-gesture classes maintain precision above 91% (Chair: 94.14%, Table: 96.79%, Fingertip: 91.84%, Finger: 97.67%). While the overall accuracy is slightly lower than in the within-user setting, the performance trends remain consistent. The Table and Finger

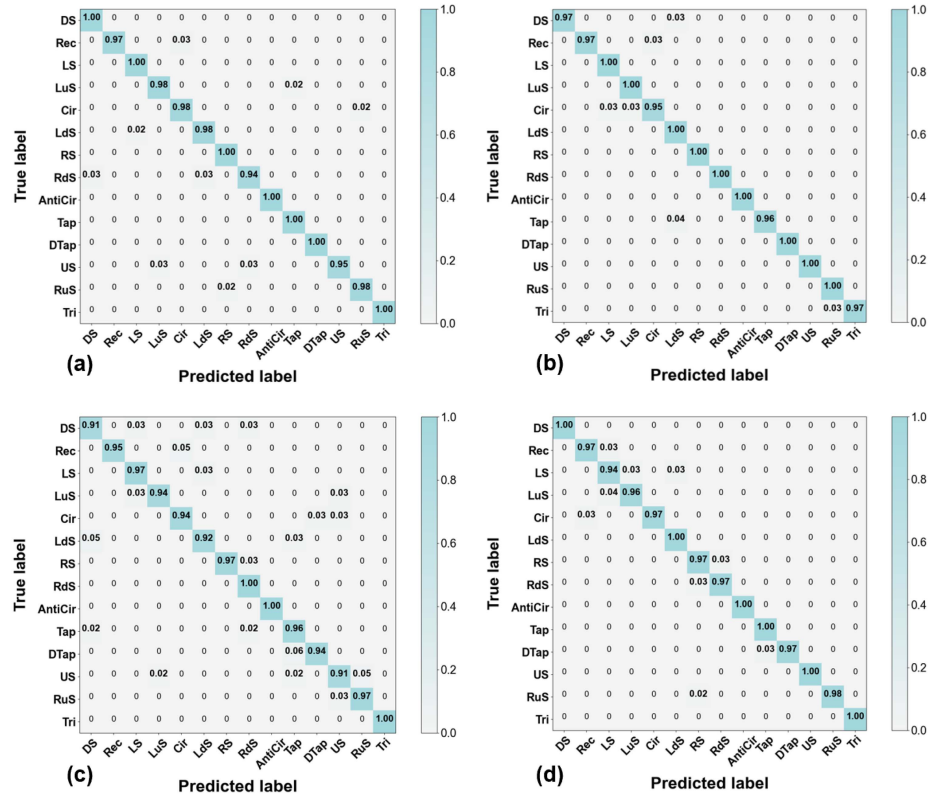


Fig. 7. Normalized confusion matrix in within-user evaluation in four positions: (a) Chair; (b) Table; (c) Fingertip; (d) Finger.

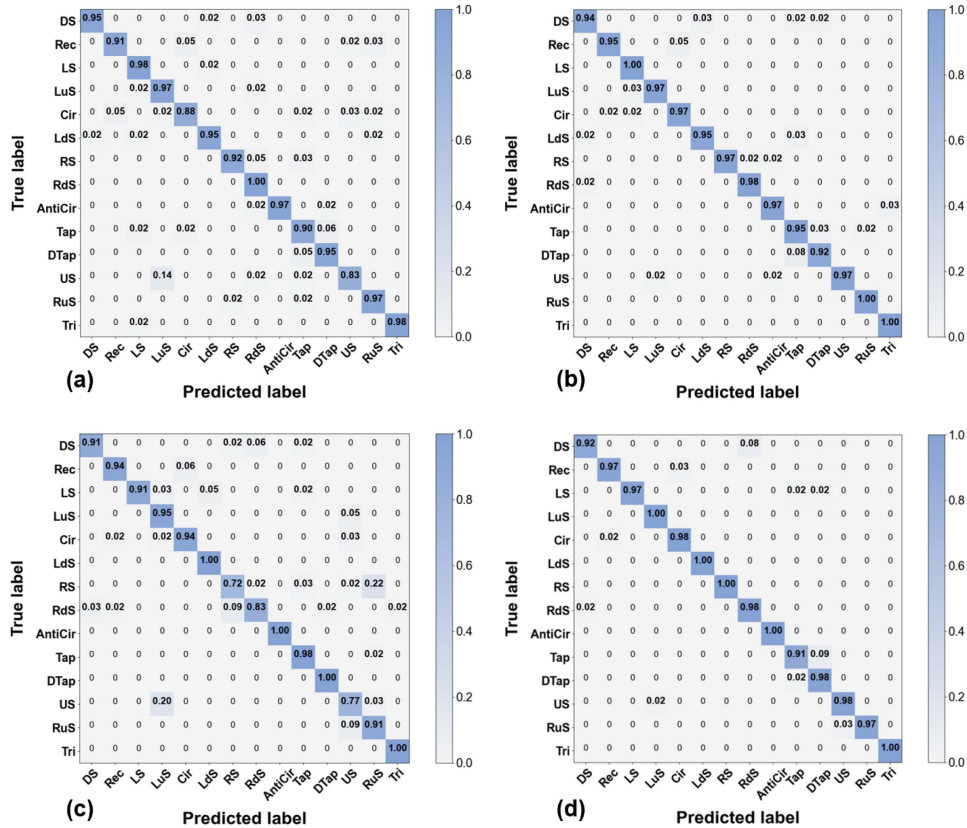


Fig. 8. Normalized confusion matrix in cross-user evaluation in four positions: (a) Chair; (b) Table; (c) Fingertip; (d) Finger.

TABLE II
TESTING ACCURACY OF EACH MICRO-GESTURE WITH DIFFERENT MODELS, INCLUDING WITHIN-USER AND CROSS-USER RESULTS

Model	Interaction position							
	Chair		Table		Fingertip		Finger	
	Within-user	Cross-user	Within-user	Cross-user	Within-user	Cross-user	Within-user	Cross-user
MLP-based	95.04%	84.97%	97.05%	91.37%	84.91%	82.58%	93.96%	88.63%
Resnet-based	97.06%	92.70%	98.15%	94.80%	93.78%	89.80%	97.09%	95.68%
DenseNet-based	96.69%	91.16%	98.15%	95.13%	94.24%	90.19%	97.39%	96.03%
VGGNet-based	97.24%	93.59%	98.24%	95.02%	94.36%	91.29%	97.64%	96.12%
Transformer-based	96.43%	90.94%	98.49%	94.03%	91.68%	86.66%	96.87%	93.24%
CNN-LSTM-based [60]	95.77%	89.06%	96.68%	92.26%	87.00%	84.23%	96.00%	90.91%
Ours+dF/dt	93.17%	90.27%	96.67%	95.74%	90.41%	88.59%	95.90%	95.64%
Ours	98.35%	94.14%	98.89%	96.79%	95.55%	91.84%	98.26%	97.67%

groups continue to outperform the Chair and Fingertip groups, and similar error patterns are observed, including confusion between Rec and Cir gestures and between adjacent-direction slide gestures. Notably, in the Fingertip group, rightward (RS) and upward (US) slide gestures exhibit relatively lower accuracy, indicating higher sensitivity to subtle directional variations in fingertip-based interactions.

E. Discussion

Integration of the pressure array sensor: The results of both the within-user and cross-user tests indicate that the Chair, Table, and Finger groups exhibit relatively high prediction accuracy, whereas the Fingertip group demonstrates comparatively lower performance. Notably, in the cross-user test, the prediction accuracy for the Fingertip group declines more significantly. This may be attributed to the soft materials used in the construction of the fingertip covers, which were chosen to enhance user comfort. However, this softness introduces increased noise in the pressure sensor readings. Additionally, variations in finger shape among different users lead to different deformation patterns in the pressure sensor array, resulting in inconsistencies in the noise distribution. These factors collectively contribute to the reduced prediction accuracy. Therefore, for practical applications, we recommend adopting alternative interaction modalities in which the sensors are mounted on rigid surfaces, such as being embedded directly into rings or small boxes. These configurations offer more stable sensing conditions and are better suited for consistent interaction. Among all configurations, the Finger and Table groups maintained relatively high accuracy in the cross-user test, indicating greater potential for generalizability. In particular, the ring-based interaction approach could be further integrated into existing wearable ring devices, as seen in some commercial products [19], [20].

The impact of micro-gesture types: Within the selected gesture set, due to the size constraints of the pressure array sensor and the limited surface area of the finger, the Rec micro-gesture is easily confused with the Cir micro-gesture. Therefore, it is advisable not to include both gestures simultaneously in the same interaction set. In contrast, the AntiCir gesture, which involves a sliding motion in the opposite direction to the aforementioned two, demonstrates higher prediction accuracy. Although the Tri micro-gesture shares a similar sliding direction

with Rec and Cir micro-gestures, it exhibits distinct motion characteristics, resulting in relatively high prediction accuracy as well. In the case of sliding gestures across eight directions, adjacent directions tend to produce a certain degree of confusion. For interaction scenarios that do not require all eight directional gestures, selecting four non-adjacent directions—such as orthogonal or diagonal directions—can lead to improved prediction performance. Additionally, while both Tap and DTap micro-gestures achieve relatively high accuracy, some confusion may still occur between them. Our analysis suggests that this may stem from users not fully lifting their fingers between taps during a double tap, which causes signal overlap. This should be carefully considered in future interaction design. **Gesture set design.** Based on the experimental results, the micro-gesture set can be further optimized in future applications to balance gesture richness and recognition robustness on a compact pressure input surface. A simplified gesture set consisting of Tap, four-way slides (US/DS/LS/RS), Circle Clockwise (Cir), and Circle Counterclockwise (AntiCir), and Triangle can be formed. This design reduces the risk of gesture confusion while retaining sufficient interactive expressiveness, providing a workable design reference for pressure-sensing wearable micro-gesture input.

V. PORTINPUT: REAL-TIME INTERACTION SYSTEM

We built a real-time gesture interaction system, *PortInput*, based on the aforementioned micro-gesture recognition model. It can recognize both the active state and idle state of gestures and perform real-time micro-gesture recognition, as shown in Fig. 9. First, *PortInput* collects raw data, which is transmitted in real time to the micro-gesture recognition processor via Bluetooth. The real-time recognition algorithm, developed in Python, then infers the interaction gestures in real time. The recognized gesture signals are subsequently transmitted to the HMD via the Transmission Control Protocol (TCP). Finally, the virtual environment processor receives the interaction signals, updates the virtual environment accordingly, and renders the results in real time on the HMD.

Algorithm 1 shows how to recognize interactive micro-gestures in real time. *PortInput* detects user input signals in real time at a frame rate of 60 Hz. When 5 channels of the 16-channel sensor detect pressure signals, they are considered valid signals.

Algorithm 1: Real-Time Micro-Gesture Recognition.

Require: Pressure sensor array S (16 channels), Prediction model f_θ
Ensure: Recognized gesture type g

```

1: Initialize states:
2:    $buffer \leftarrow \emptyset$   $\triangleright$  Data buffer
3:    $counter \leftarrow 0$   $\triangleright$  Valid frame counter (threshold=3)
4:    $t_{start} \leftarrow \text{null}$   $\triangleright$  Segment start time
5:    $t_{last\_valid} \leftarrow \text{null}$   $\triangleright$  Last valid frame timestamp
6: while true do  $\triangleright$  Main processing loop
7:    $(x_t, t) \leftarrow \text{READFRAME}()$   $\triangleright$  Unified frame reading
8:   if  $t_{start} = \text{null}$  then
9:     if  $\text{VALIDSIGNAL}(x_t)$  then
10:      Initialize first segment:
11:       $t_{start} \leftarrow t, t_{last\_valid} \leftarrow t, buffer \leftarrow \{(t, x_t)\},$ 
12:       $counter \leftarrow 1$ 
13:    end if
14:  else
15:    if  $\text{VALIDSIGNAL}(x_t)$  then
16:       $buffer \leftarrow buffer \cup \{(t, x_t)\}, counter \leftarrow$ 
17:       $counter + 1$ 
18:       $t_{last\_valid} \leftarrow t$ 
19:    else
20:      if  $counter > 3$  then  $\triangleright$  Minimum valid frames
21:        if  $\text{ISTIMEOUT}(t, t_{last\_valid}, 0.2)$  then
22:           $g \leftarrow \text{PREDICTGESTURE}(buffer, t_{start}, t_{last\_valid})$ 
23:           $\text{RESETSTATES}()$ 
24:          return  $g$ 
25:        else
26:           $g \leftarrow \text{MONITORNEXTSEGMENT}()$ 
27:           $\text{RESETSTATES}()$ 
28:          return  $g$ 
29:        end if
30:      end if
31:    end if
32:  end while
33: function  $\text{PREDICTGESTURE}(buf, t_s, t_e)$   $\triangleright$  Unified prediction
34:    $raw\_data \leftarrow \{(t, f) \in buf \mid t_s \leq t \leq t_e\}$ 
35:    $N \leftarrow |raw\_data|$   $\triangleright$  Raw frame count
36:   if  $N < 64$  then
37:     Upsample each channel with cubic spline interpolation
38:   else if  $N > 64$  then
39:     Downsample each channel with average pooling
40:   else
41:     Directly use original 64 frames
42:   end if
43:   Construct tensor  $X \in \mathbb{R}^{16 \times 64}$ 
44:   Maximum-minimum normalization  $X$  to the  $[0, 1]$ 
45:   return  $f_\theta(X)$ 
46: function  $\text{READFRAME}$   $\triangleright$  Unified sensor reading

```

```

47:    $x_t \leftarrow \text{ReadSensor}(S)$   $\triangleright$  60Hz sampling
48:    $t \leftarrow \text{GetTimestamp}()$ 
49:   return  $(x_t, t)$ 
50: end function
51: function  $\text{MONITORNEXTSEGMENT}$   $\triangleright$  Sub-segment monitoring
52:   while true do
53:      $(x_t, t) \leftarrow \text{READFRAME}()$ 
54:     if  $\text{VALIDSIGNAL}(x_t)$  then
55:        $buffer \leftarrow buffer \cup \{(t, x_t)\}$ 
56:       return  $\text{PREDICTGESTURE}(buffer, t_{start}, t)$ 
57:     else if  $\text{ISTIMEOUT}(t, t_{last\_valid}, 0.2)$  then
58:       return  $\text{PREDICTGESTURE}(buffer, t_{start}, t_{last\_valid})$ 
59:     end if
60:   end while
61: end function
62: function  $\text{ISTIMEOUT}(t_1, t_2, \tau)$   $\triangleright$  Timeout check
63:   return  $(t_1 - t_2) \geq \tau$ 
64: end function
65: function  $\text{RESETSTATES}$   $\triangleright$  State cleanup
66:    $buffer \leftarrow \emptyset, counter \leftarrow 0$ 
67:    $t_{start} \leftarrow \text{null}, t_{last\_valid} \leftarrow \text{null}$ 
68: end function

```

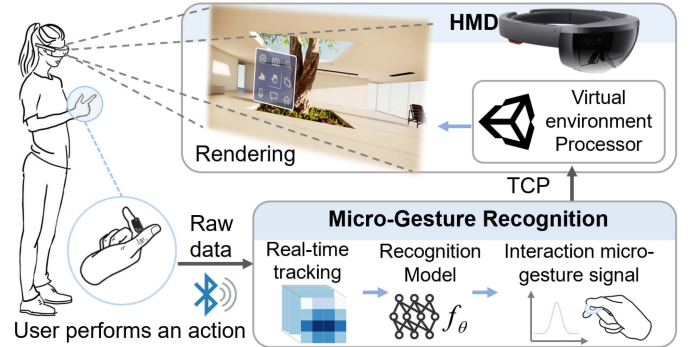


Fig. 9. System Workflow of *PortInput*. Raw sensor data is first collected and transmitted via Bluetooth to the micro-gesture recognition processor in real time at 60 Hz. Then, a deep learning-based recognition algorithm infers the interaction micro-gestures in real time. The recognized gesture signals are subsequently sent to the HMD using the TCP protocol. Finally, the virtual environment processor on the HMD, developed using Unity, receives the interaction commands and renders the corresponding changes on the HMD in real time.

First, when the sensor detects valid signals for 3 consecutive frames, the system determines that the input action has started and initializes the data buffer, and continues to capture valid frames until invalid signals appear. When invalid signals appear, the algorithm makes dynamic decisions through a dual timing detection mechanism - if no new signal appears within 0.2 seconds after the last valid frame, the current action is determined to be an independent segment; if a valid signal is detected again within the 0.2 s timeout window, the sub-segment monitoring mode is started to capture potential continuous actions, thereby realizing the detection of double-click actions. For the captured

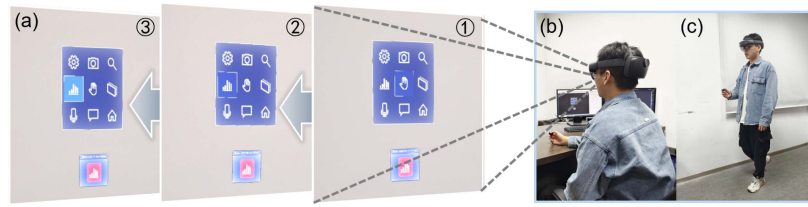


Fig. 10. (a) Interaction layout for selecting tasks. Step ①: The target icon to be selected is displayed at the bottom, while the default candidate icon is positioned at the top center, visually indicated by a semi-transparent blue highlight. Step ②: The participant selects another candidate icon by performing a slide micro-gesture in one of eight directions (*PortInput*) or by pointing with a ray (*Pointing-Pinch*). Step ③: The participant confirms the selection via a Tap micro-gesture (*PortInput*), a Pinch gesture (*Pointing-Pinch*), or by directly clicking with the index finger (*Air-Tap*). The selected icon is then visually highlighted in solid blue as feedback. (b) Experimental setup under the *sitting* condition; (c) Experimental setup under the *walking* condition.

raw data stream, the system adopts an adaptive normalization strategy to normalize the 16-channel data to the [0,1] interval through maximum-minimum value mapping. Then, resampling technology (cubic spline interpolation to make up 64 frames or average pooling to compress very long sequences) is used to ensure that the data input to the model is always 64 frames long. Finally, the preprocessed multi-dimensional time series features are input into the prediction model for end-to-end micro-gesture classification. The algorithm achieves robust capture of intermittent gestures while ensuring real-time performance.

VI. USER STUDY 1 - COMPARISON WITH TRADITIONAL METHODS IN AR HMDs

In this section, we conduct a user study to evaluate the performance of PortInput. Since *PortInput* can support users in different physical states, we assess its performance under both *sitting* and *walking* conditions. We select the **Finger** prototype for comparison. In addition, current commercial HMDs commonly support two bare-hand interaction methods in these states: (1) the *Pointing-Pinch* method, where users point at a target using raycasting and confirm selection with a pinch gesture, and (2) the *Air-Tap* method, where users directly tap on virtual objects using the index finger within tracked hands [61], [62].

A. Participants and Apparatus

20 new participants (11 males, 9 females) were recruited for this study, aged 21-31 years ($M = 25.3$, $SD = 6.37$). All participants are right-handed and performed all tasks using their right hand. Among them, only 4 participants have limited experience with AR/VR HMDs, while the remaining participants have never used such devices. None of the participants has any prior experience with micro-gesture interaction devices. The experiment has obtained the informed consent of the participants. The experiment is conducted using Microsoft HoloLens 2. The micro-gesture recognition algorithm is run on a system equipped with an Intel Core i7 processor and an NVIDIA RTX 3070Ti graphics card. The software is implemented using C# in Unity2023.3.45f1c1 and Python3.9.20.

B. Task and Procedure

This study adopts a within-subjects design and employs a 2 (user states) $\times$ 3 (interaction methods) two-factor setup. The

user state refers to the participant's physical condition—either *walking* or static *sitting*, as shown in Fig. 10. The interaction method includes three input techniques: *PortInput*, *Pointing-Pinch*, and *Air-Tap*, resulting in a total of six experimental groups. In addition to the above two influencing factors, users perform the same selection tasks in each group.

Fig. 10(a) illustrates the interaction layout for the selection task. The target icon is displayed at the bottom of the interface. In each trial, the target icon is first presented. In the *PortInput* method, the default candidate icon (highlighted with a semi-transparent box) initially appears in the center-top position (step ① in Fig. 10(a)). Participants use slide-based micro-gestures in eight directions to navigate among candidate icons (step ②), and confirm their selection with a Tap micro-gesture. Once selected, the icon is highlighted in blue as visual feedback (step ③). In the *Pointing-Pinch* method, the participant points at the desired icon with a virtual ray to designate it as the candidate, and then confirms the selection with a pinch gesture. In the *Air-Tap* method, the participant directly selects the target icon by tapping it with their index finger within the tracked hand.

Following HoloLens interaction design guidelines [63], in both the *PortInput* and *Pointing-Pinch* methods, icons are positioned 2 meters in front of the user at the center of the field of view, with each icon sized at $3.5 \text{ cm} \times 3.5 \text{ cm}$. In the *Air-Tap* method, icons are placed 45 cm in front of the participant, also centered in the field of view, and each icon is sized at $1.6 \text{ cm} \times 1.6 \text{ cm}$. In the *walking* condition, participants complete the tasks while walking along a predefined route shown in Fig. 11. The experiment is conducted in a $6.5 \text{ m} \times 4 \text{ m}$ room with tables and chairs, requiring participants to walk at a normal pace while avoiding obstacles.

Before the experiment, participants are guided by the experimenter to learn and become familiar with the selection method for each condition with several training trials. Each participant completes all six groups in six sessions, with the order of groups counterbalanced using a Latin square design. Participants are allowed to rest between sessions. During the experiment, they are instructed to perform the tasks as quickly as possible. After each session, participants complete a questionnaire consisting of a NASA-TLX [64] (7-point Likert scale), the System Usability Scale (SUS) [65], and the short version of the User Experience Questionnaire (UEQ-S) [66] on a 5-point Likert scale and give their other feedback.

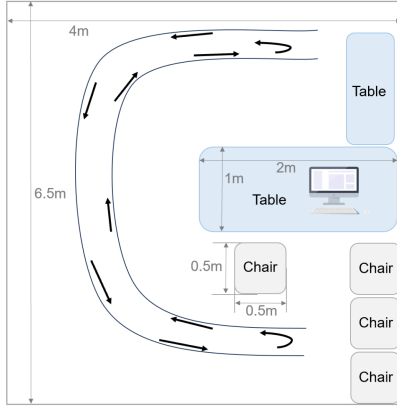
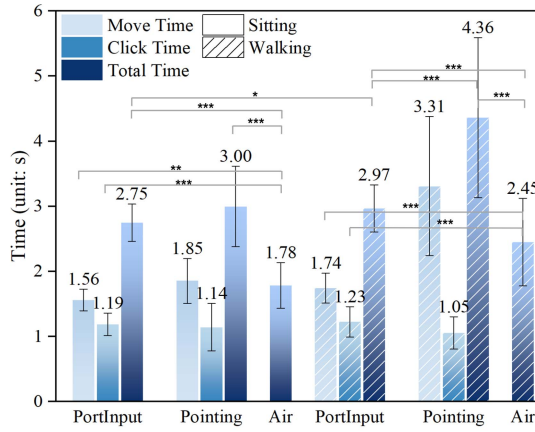


Fig. 11. Walking route.

Fig. 12. Average time. Error bars indicate ± 1 standard deviation. (*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).

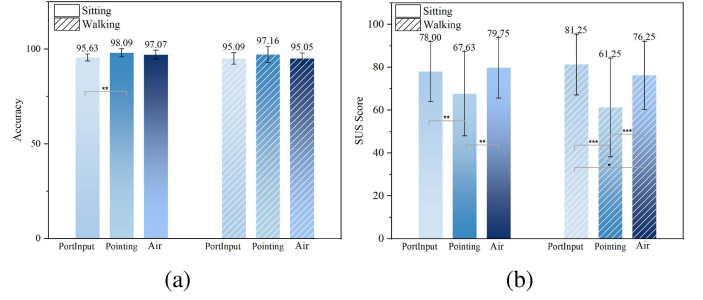
In total, we collect 2 (user states) $\times 3$ (interaction methods) $\times 20$ (participants) $\times 10$ (repeated) = 1,200 trials.

C. Results

We conducted normality testing using the Shapiro–Wilk test. Parametric tests (paired t-tests, repeated-measures ANOVA) were applied when assumptions were met, with corrections for sphericity violations, and non-parametric tests (Wilcoxon signed-rank, Friedman) were used otherwise. All analyses were performed using IBM SPSS Statistics 25.

Time: Fig. 12 shows task completion times under two conditions. The *Pointing-Pinch* group is denoted as **Pointing**, and the *Air-Tap* group is denoted as **Air** in all subsequent figures. For the *PortInput* and *Pointing-Pinch* groups, each interaction consists of two stages: selecting a candidate and confirming the selection. Therefore, we measured both the total completion time and the time spent on each stage. The time taken to select a candidate is referred to as **Move Time**, while the time required to confirm the selection is referred to as **Click Time**.

Task completion times were significantly shorter under the *sitting* condition than *walking* for all input methods (*PortInput*: $p = 0.014$; *Pointing-Pinch*: $p < 0.001$; *Air-Tap*: $p = 0.001$). A

Fig. 13. (a) Average accuracy (%). (b) Average scores of the SUS questionnaire. Error bars indicate ± 1 standard deviation. (*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).

significant difference was observed among input methods under both sitting ($p < 0.001$) and walking ($p < 0.001$) conditions. Post-hoc tests showed that *Air-Tap* was fastest, *Pointing-Pinch* slowest, and *PortInput* intermediate. Notably, when considering single-gesture completion (**Move Time**), *PortInput* outperformed *Air-Tap* significantly in both sitting ($p = 0.005$) and walking ($p < 0.001$), indicating its potential for efficient single-gesture interactions.

Accuracy: Fig. 13(a) shows task accuracy for all groups. Under the *sitting* condition, a significant difference was found among the three input methods ($F_{2,38} = 6.422$, $p = 0.004$). *PortInput* (95.63%) was slightly lower than *Pointing-Pinch* (98.09%, $p = 0.001$) but comparable to *Air-Tap* (97.07%, $p = 0.056$). Under the *walking* condition, no significant differences were observed among the three methods ($F_{2,38} = 2.655$, $p = 0.083$), and *PortInput* maintained similar performance between sitting and walking ($p = 0.454$). These results indicate that *PortInput* provides stable and reliable gesture recognition across different contexts.

SUS: Fig. 13(b) shows SUS results, processed following [65]. Scores differed significantly among input methods in sitting ($F_{1,473,27.983} = 21.549$, $p = 0.001$) and walking ($F_{1,379,26.193} = 21.549$, $p < 0.001$). *PortInput* and *Air-Tap* outperformed *Pointing-Pinch*, with *PortInput* highest in walking ($t_{19} = 2.486$, $p = 0.022$). No difference was found for *PortInput* between sitting and walking.

UEQ-S: Fig. 14 shows the UEQ-S results, processed according to [67]. **Pragmatic Quality.** Significant group differences were found in sitting ($F_{2,38} = 21.549$, $p = 0.002$) and walking ($F_{1,345,25.547} = 15.694$, $p < 0.001$), with *PortInput* and *Air-Tap* outperforming *Pointing-Pinch*. *PortInput* scored highest in walking and was stable across conditions ($t_{19} = 0.733$, $p = 0.473$). **Hedonic Quality.** Differences were significant in sitting ($F_{2,38} = 3.566$, $p = 0.039$) and walking ($F_{1,498,28.466} = 5.663$, $p = 0.014$). *PortInput* consistently outperformed the others. **Overall.** Significant differences were observed in sitting ($F_{1,428,27.130} = 6.169$, $p = 0.005$) and walking ($F_{1,530,29.069} = 13.184$, $p < 0.001$), with *PortInput* higher than *Pointing-Pinch* and *Air-Tap* in both situations.

Workload: Fig. 15 shows NASA-TLX results. In *sitting*, only the *performance* dimension did not show a significant difference across the three techniques ($\chi^2_{20} = 0.776$, $p = 0.679$),

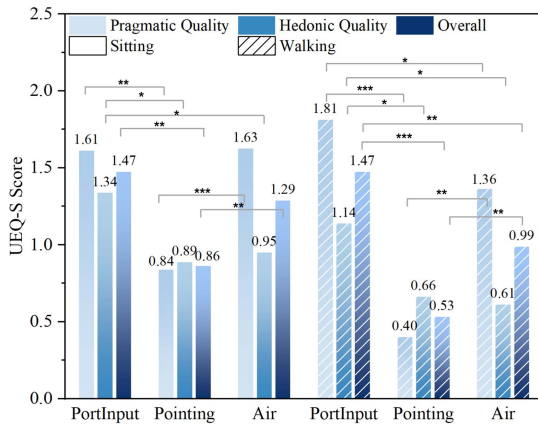


Fig. 14. Average scores of UEQ-S questionnaire. Error bars indicate ± 1 standard deviation. (*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).

while other subscales all showed significant differences. In the *walking* condition, no significant differences were observed in *temporal demand* ($\chi^2_{20} = 2.625$, $p = 0.269$) and *performance* ($\chi^2_{20} = 4.719$, $p = 0.094$), while other subscales all with significant difference. Across all subscales, *PortInput* scores were stable between sitting and walking, indicating robust workload performance.

User feedback: After the experiment, we collected the feedback of users. For the selection of nearby icons, most participants preferred to use the *Air-Tap* method when sitting, and a few participants preferred *PortInput*; when walking, participants preferred to use *PortInput* to select rather than the *Air-Tap* method. For the selection of distant icons, participants preferred *PortInput* rather than *Pointing-Pinch* regardless of whether they were sitting or walking. For *PortInput*, participants felt that the operation was simple and clear and easy to understand; participants felt that *PortInput* could maintain a similar selection accuracy without being affected by the state when walking, and allowed people to interact while walking, which was suitable for applications in non-sitting states, such as interactions in games. For the other two methods, some participants felt that their arms were always hanging in the air, and their arms got tired.

D. Discussion

Performance in sitting and walking conditions: The experimental results indicate that *PortInput* achieves shorter task completion times under the *sitting* condition compared to the *walking* condition, while maintaining similar accuracy across both contexts. Execution times for individual gestures are also shorter in the *sitting* condition. However, subjective evaluation, including the SUS, the UEQ-S, and NASA-TLX workload questionnaires, shows no significant differences in users' perceived experience between the two conditions. Interestingly, users rate the *practicality* of *PortInput* higher under the *walking* condition. Overall, the consistent performance of *PortInput* across both scenarios suggests its potential for broad applicability in diverse mobile and stationary contexts.

Performance compared with Pointing-Pinch : In terms of task completion time, *PortInput* outperforms the *Pointing-Pinch* technique in both *sitting* and *walking* conditions, with a more pronounced difference observed while walking. Although the accuracy of *PortInput* is slightly lower than that of *Pointing-Pinch* in both settings, this may be attributed to the instruction for participants to prioritize speed over precision; errors were not corrected during the experiment. In practical usage, users could employ slide gestures for error correction. From a user experience perspective, *PortInput* demonstrates superior usability (SUS) and user experience (UEQ-S) across pragmatic, hedonic, and overall dimensions. It also imposes a lower workload (NASA-TLX) compared to the *Pointing-Pinch* method. These findings demonstrate that *PortInput* delivers better performance and user satisfaction than the traditional *Pointing-Pinch* approach in both static and mobile conditions.

Performance compared with Air-Tap : *PortInput*'s dual-gesture mechanism (directional slide and tap) results in slightly longer task completion times than *Air-Tap*, but both methods achieve comparable accuracy. In the future, this could be achieved for better performance by using a single gesture for selection. Usability scores (SUS) are similar while sitting, and *PortInput* is rated higher while walking. In terms of user experience (UEQ-S), *PortInput* matches *Air-Tap* in pragmatic quality and exceeds it in hedonic quality when sitting, and outperforms it in both dimensions while walking. Regarding workload, *PortInput* imposes higher mental and temporal demand but lower physical demand, reflecting the ergonomic advantage of micro-gestures over prolonged hand elevation in *Air-Tap*. Although *Air-Tap* achieves faster input in the current task, it may induce physical fatigue. Notably, single-gesture interactions in *PortInput* can be faster, suggesting potential for improved efficiency in future designs [5], [28]. Overall, while *Air-Tap* shows higher input efficiency in the current setup, *PortInput* offers a better user experience and has great potential for optimization and real-world application.

VII. USER STUDY 2 - COMPARISON WITH THE COMMERCIAL PRESSURE-BASED RING

To further verify the performance of *PortInput*, we conducted this experiment and compared it with an existing commercial pressure-based device, Genius Ring [47].

A. Participants and Apparatus

16 new participants (9 males, 7 females) were recruited for this study, aged 21-27 years ($M = 23.81$, $SD = 1.33$). 10 participants had limited experience (used occasionally), while the other 6 had regular experience with AR/VR headsets. They all used their dominant hand (right hand) to wear the device during the experiment. The experiment has obtained the informed consent of the participants. In this study, we use the Genius Ring while the other apparatuses are the same as in study 1. Genius Ring is a ring-shaped controller with a black circular touch-sensitive area at its center, whose primary function is to enable cursor control by touching and sliding on the black circular touch zone, and

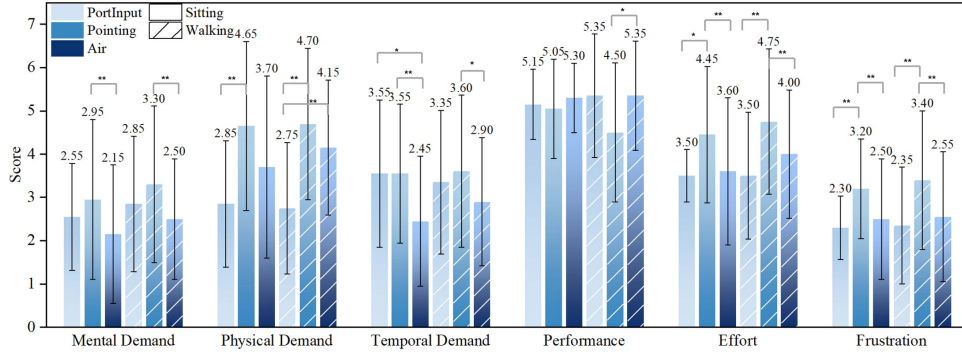


Fig. 15. Average scores of NASA-TLX questionnaire. Error bars indicate ± 1 standard deviation. (*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).

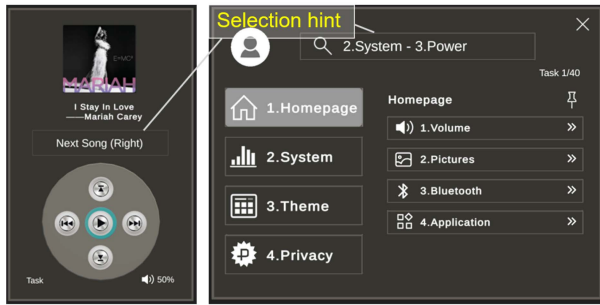


Fig. 16. Illustration of tasks in Study 2. Left: Music Menu; Right: Hierarchical Menu.

when the cursor enters the preset target range, pressing the touch zone completes target selection.

B. Task

We design two tasks as shown in Fig. 16. **Music Menu**, which simulates a media player interface with five discrete commands: “Next” (right), “Last” (left), “Volume up” (up), “Volume down” (down), and “Toggle” (center). **Hierarchical Menu**, which requires navigating a two-level menu consisting of four parent categories (labeled 1–4), each containing four child items, resulting in 16 leaf nodes. Parent categories are displayed on the left, with corresponding child items shown on the right, while the child-level menu is dynamically updated based on the selected parent.

Both tasks are presented as spatially anchored 2D interfaces in the AR HMD. In **Music Menu** task, the music player interface is displayed as a $0.48 \text{ m} \times 0.80 \text{ m}$ rectangular panel at a fixed viewing distance of 3 m, with each icon measuring $0.064 \text{ m} \times 0.064 \text{ m}$ and spaced 0.07 m apart. The larger viewing distance is selected to accommodate the overall panel size while maintaining comfortable visual coverage within the user’s field of view. In **Hierarchical Menu** task, the menu occupies an area of $0.60 \text{ m} \times 0.45 \text{ m}$ at a fixed viewing distance of 2 m. Each parent icon measures $0.21 \text{ m} \times 0.054 \text{ m}$ with 0.021 m spacing, while each child icon measures $0.30 \text{ m} \times 0.032 \text{ m}$ with 0.016 m spacing. Across both tasks, all icons exceed the minimum recommended target size specified in the Microsoft

HoloLens design guidelines [63], ensuring reliable selection and visual legibility.

As shown in Fig. 16, participants select the target icon according to a displayed selection hint. In both tasks, when using the *Genius Ring*, users continuously slid to move the cursor within the target icon (highlighted in green or indicated by a gray background) and then click to confirm the selection. When using *PortInput*, to ensure a fair comparison with the *Genius Ring*, a similar two-step selection procedure is adopted: users first perform a micro-gesture to select the target icon and then execute a *Tap* micro-gesture to confirm the selection. In future practical applications, icons could be directly selected using gestures without an explicit confirmation step, as demonstrated in the *MouseRing* [46]. In the **Music Menu** task, directional gestures are mapped to the corresponding directional icons: **US** gesture for selecting the Up icon, **DS** for selecting the Down icon, **LS** for selecting the Left icon, and **RS** for selecting the Right icon, followed by a **Tap** to confirm the selection. In the **Hierarchical Menu** task, **US** and **DS** gestures are used to navigate between icons within the same menu level, while **LS** and **RS** gestures are used to switch between the parent (left) and child (right) levels. A **Tap** gesture is used to confirm the current selection. When a selection is confirmed in the parent level, the system automatically moves the cursor to the top icon of the child level; conversely, confirming a selection in the child level causes the cursor to return to the top icon of the parent level.

C. Procedure

This study adopts a within-subjects design and a two-factor experimental setup (4 groups) with $2 \text{ tasks} \times 2 \text{ interaction methods}$. Before the experiment, participants have five minutes to familiarize themselves with the operation of both devices. The group order is balanced using a Latin square design. Each participant completes 4 blocks in each group, with each block consisting of 10 choices. To ensure a balanced distribution of stimuli, selection hints are generated using a Fisher-Yates shuffle algorithm for the selection of all icons in both tasks. During the experiment, participants are instructed to complete the tasks as quickly as possible. After each experimental phase, participants complete the same questionnaire as in Study 1. A total of $2 \text{ (user status)} \times 2 \text{ (interaction methods)} \times 14 \text{ (participants)} \times 4$

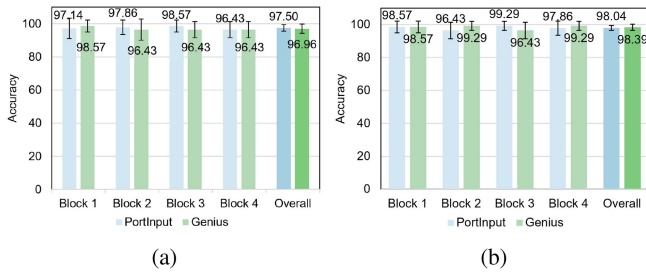


Fig. 17. Average accuracy (%) of Music Menu (a) and Hierarchical Menu (b) tasks. Error bars indicate ± 1 standard deviation.

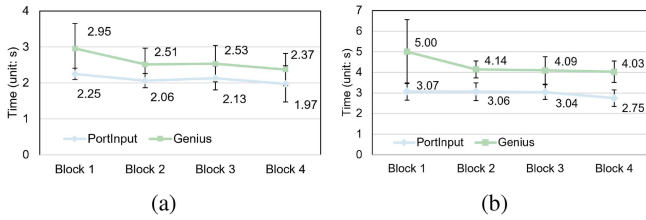


Fig. 18. Average time of Music Menu (a) and Hierarchical Menu (b) tasks. Error bars indicate ± 1 standard deviation.

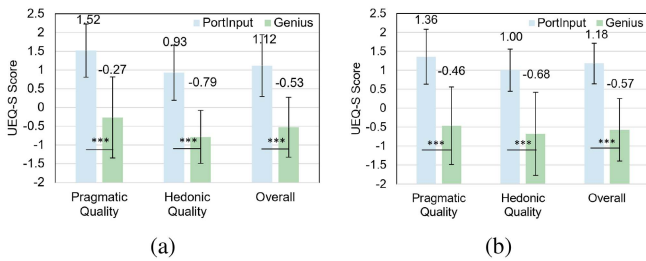


Fig. 19. Average UEQ-S score of Music Menu (a) and Hierarchical Menu (b) tasks. Error bars indicate ± 1 standard deviation. (***: $p < 0.001$).

(blocks) $\times$ 10 (repeated) = 2240 experimental data points are collected.

D. Results

We used the same data analysis methods as in Study 1.

Accuracy: Fig. 17(a) shows task completion times of two tasks across four blocks. The results show that both selection methods had high accuracy in both tasks, with no significant difference. Furthermore, the performance of both selection methods tended to stabilize across two tasks (PortInput: $F_{3,39} = 1.140$, $p = 0.345$; Genius Ring: $F_{3,39} = 2.004$, $p = 0.129$).

Time: Fig. 18(a) shows completion time. The results show that task completion time in both tasks significantly reduced across blocks in both methods (PortInput: $F_{3,39} = 5.634$, $p = 0.003$; Genius Ring: $F_{1,244,16.171} = 5.924$, $p = 0.021$). Furthermore, within each block, the completion time of PortInput was significantly shorter than that of Genius Ring.

UEQ-S: Fig. 19 shows the UEQ-S results. The results show that PortInput's perceived user satisfaction was significantly higher than that of Genius Ring in terms of Pragmatic Quality

and Hedonic Quality. **SUS.** Fig. 20(a) shows SUS results. The results show that PortInput's user-aware system usability was significantly higher than Genius Ring in both tasks. **Workload.** Fig. 20(b) shows NASA-TLX results. The results showed that, across all subscales, there was no significant difference in cognitive load between the two tasks for each selection method. Furthermore, except for the Music Menu task, where there was no significant difference in mental demand between PortInput and Genius Ring, in the other subscales, the cognitive load of PortInput was significantly lower than that of Genius Ring across both tasks.

User feedback: Qualitative feedback was collected after the experiment. Overall, participants reported a strong preference for PortInput over the Genius Ring in both tasks. Many participants described PortInput as "more intuitive" and "more natural", noting that directional micro-gestures aligned well with their mental models of menu navigation, particularly in the Music Menu task. Several participants commented that they could quickly understand and remember the gesture-command mappings, which reduced the need for visual confirmation during interaction. In contrast, while the Genius Ring was perceived as familiar and reliable, some participants reported that continuous sliding to position the cursor required sustained attention and fine motor control. Participants also highlighted that the two-step selection strategy in PortInput (gesture selection followed by a Tap confirmation) provided a sense of control and reduced accidental selections, despite adding an extra action. Several participants noted that they could imagine using PortInput with direct gesture-based selection (without confirmation) in real-world scenarios once they became more confident with the system.

E. Discussion

Performance: The results demonstrate that PortInput provides an effective alternative to a commercial pressure-based ring device for AR selection interaction. While both interaction methods achieved similarly high accuracy across tasks, PortInput consistently resulted in shorter task completion times and higher subjective usability and experience scores. These findings suggest that micro-gesture-based input can improve interaction efficiency without sacrificing precision. Across blocks, both techniques showed reduced completion time, indicating a general learning effect; however, PortInput consistently achieved shorter times within each block, suggesting faster performance stabilization. Although a two-step selection procedure was adopted for PortInput to ensure a fair comparison, users still completed tasks faster than with the Genius Ring. Moreover, user feedback indicates that removing the confirmation step in future designs—similar to approaches used in prior work such as MouseRing—could further improve performance and perceived fluidity in practical applications.

Advantages of gesture mapping: One key factor contributing to PortInput's performance advantage is the direct mapping between gesture direction and menu semantics. Unlike cursor-based interaction with the Genius Ring, which requires continuous visual monitoring and fine-grained motor control,

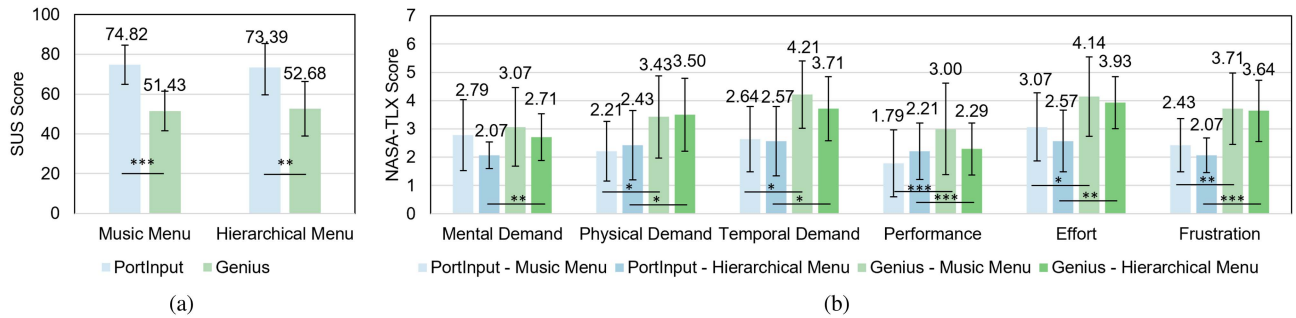


Fig. 20. (a) Average SUS score. (b) Average NASA-TLX score. Error bars indicate ± 1 standard deviation. (*: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).

PortInput allows users to issue discrete, intention-driven commands. This design reduces visuomotor coupling demands and may explain the significantly lower workload ratings observed in most NASA-TLX subscales. The benefit was particularly evident in the **Hierarchical Menu** task, where frequent level switching amplified the cost of cursor repositioning for the ring-based method.

Finally, the higher UEQ-S and SUS scores indicate that PortInput not only improves objective performance but also enhances overall user experience. The higher hedonic quality ratings suggest that gesture-based input may feel more engaging and enjoyable, which is an important consideration for prolonged AR/VR use. Together, these findings highlight the potential of micro-gesture-based ring input as a lightweight, low-effort interaction technique for AR/VR interaction.

VIII. LIMITATION AND FUTURE WORK

First, although the proposed recognition model demonstrates high cross-user accuracy across all prototypes, the current system does not incorporate personalization mechanisms. As observed in both user studies, participants were able to achieve stable performance after brief practice; nevertheless, future work could further improve accuracy and interaction efficiency by incorporating a small amount of user-specific data to personalize the micro-gesture model. Second, while Study 1 evaluated interaction performance under both *sitting* and *walking* conditions, the walking scenario was conducted in a controlled laboratory environment. Real-world mobile use of *PortInput*, particularly in public or outdoor settings, may introduce additional challenges such as uneven terrain or environmental distractions. Evaluating robustness in more naturalistic environments remains an important direction for future work. Third, to ensure fair comparisons with mid-air techniques and a commercial pressure-based ring device in Study 2, the current task design adopted a two-step selection strategy. Although this design improved reliability, it also increased task completion time compared to single-step techniques such as *Air-Tap*. Future interaction designs should prioritize *single-gesture* input strategies [5], [28], which may further enhance efficiency in practical AR menu interactions. Finally, our user studies focused on AR-based tasks. Additional application scenarios,

such as smart home control, in-vehicle interaction, and fully eyes-free command invocation (e.g., gesture-based wake-up), remain to be explored to further validate the generalizability of *PortInput*.

IX. CONCLUSION

In this work, we first introduce three pressure array sensor-based prototypes—finger-cot, finger-worn, and surface-based—that enable micro-gesture interaction on the fingertip, finger, tabletop, and chair armrest. We then propose a deep neural network-based gesture recognition algorithm capable of recognizing 14 micro-gestures across these prototypes. The within-user recognition accuracy exceeds 95%, while cross-user accuracy remains above 91%. Further analysis reveals that certain gestures achieve significantly higher recognition rates, suggesting that future gesture design should prioritize gestures with high accuracy and low confusion. Then, we develop *PortInput*, a real-time interaction system that enables continuous tracking and recognition of micro-gestures in AR HMDs. Both the hardware prototypes and the underlying algorithm are extensible to other interactive platforms, such as smart TVs and home automation systems.

We conducted two user studies to evaluate the performance of *PortInput*. The first study examines an icon selection task under both *sitting* and *walking* conditions, comparing *PortInput* with widely used mid-air interaction techniques in AR, including *Pointing-Pinch* and *Air-Tap*. The results show that *PortInput* outperforms *Pointing-Pinch* in both task performance and user experience across conditions, with more pronounced advantages during walking. The second study compares *PortInput* with a commercial pressure-based ring device in flat and hierarchical AR menu selection tasks. The results show that *PortInput* achieves comparable accuracy while enabling faster task completion, lower perceived workload, and higher subjective usability and user experience, highlighting its effectiveness relative to existing commercial solutions.

Overall, *PortInput* demonstrates applicability across diverse interaction contexts. It also highlights the promising potential of integrating pressure array sensors into existing ring-based devices [19], [20] and other wearable platforms, laying the foundation for intuitive and efficient interaction modes.

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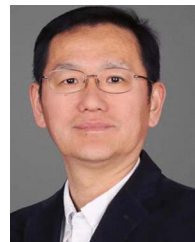
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